



## AQ5.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

Let  $T(x, y, z) = mx + ny + pz$  denote the total amount of money a multiplex made from a specific movie, in one day by selling  $x$  number of tickets for the morning show,  $y$  number of tickets for the evening show, and  $z$  number of tickets for the night show, where  $m$ ,  $n$  and  $p$  are the prices of each ticket of that movie in the morning, evening, and night show, respectively.

In the multiplex, 2 Malayalam movies, 1 Hindi movie, 1 Bengali movie, and 1 English movie are running parallelly in one week. Suppose there are 3 shows (Morning, Evening, Night) per day for each of the Malayalam movies and the English movie, whereas 2 shows (Evening, Night) per day for the Hindi and the Bengali movie. Ticket price (irrespective of morning, evening, and night shows) for each Malayalam movie is ₹200, Hindi movie is ₹250, Bengali movie is ₹250, and English movie is ₹300. The number of tickets sold in a particular day is given in Table M2W5AQ2:

Answer questions 1 to 8 using the given data above.

### Level 1

1 point

Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the morning shows only?



$$200m_1 + 200m_2 + 200m_3 \quad 200m_1 + 200m_2 + 200m_3$$



$$200m_1 + 250m'_1 + 300e_1 \quad 200m_1 + 250m'_1 + 300e_1$$



$$200m_1 + 200m'_1 + 250e_1 \quad 200m_1 + 200m'_1 + 250e_1$$



$$200m_1 + 200m'_1 + 300e_1 \quad 200m_1 + 200m'_1 + 300e_1$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$200m_1 + 200m'_1 + 300e_1 \quad 200m_1 + 200m'_1 + 300e_1$$

1 point

Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Bengali movie only?



$$200(b_2 + b_3) \quad 200(b_2 + b_3)$$



$$250(b_2 + b_3) \quad 250(b_2 + b_3)$$



$$250(h_2 + h_3) \quad 250(h_2 + h_3)$$



$$250b_2 + 200b_3 \quad 250b_2 + 200b_3$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**



$$250(b_2 + b_3)250(b_2 + b_3)$$

1 point

Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the night shows only?

☐

$$200(m_2 + m'_2) + 250(h_2 + b_2) + 300e_2 \quad 200(m_2 + m'_2) + 250(h_2 + b_2) + 300e_2$$

☐

$$200(m_3 + m'_3 + h_3 + b_3) + 300e_3 \quad 200(m_3 + m'_3 + h_3 + b_3) + 300e_3$$

☐

$$200(m_3 + m'_3) + 250(h_3 + b_3) + 300e_3 \quad 200(m_3 + m'_3) + 250(h_3 + b_3) + 300e_3$$

☐

$$200(m_3 m'_3) + 250(h_3 b_3) + 300e_3 \quad 200(m_3 m'_3) + 250(h_3 b_3) + 300e_3$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$200(m_3 + m'_3) + 250(h_3 + b_3) + 300e_3 \quad 200(m_3 + m'_3) + 250(h_3 + b_3) + 300e_3$$

1 point

Which expression accurately expresses the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Bengali and English movie only?

☐

$$250(b_2 + b_3)250(b_2 + b_3)$$

☐

$$300e_1 + 250(b_2 + e_2) + 300(b_3 + e_3) \quad 300e_1 + 250(b_2 + e_2) + 300(b_3 + e_3)$$

☐

$$250(b_2 + b_3) + 300(e_1 + e_2 + e_3) \quad 250(b_2 + b_3) + 300(e_1 + e_2 + e_3)$$

☐

$$b_2 + b_3 + e_1 + e_2 + e_3 \quad b_2 + b_3 + e_1 + e_2 + e_3$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$250(b_2 + b_3) + 300(e_1 + e_2 + e_3) \quad 250(b_2 + b_3) + 300(e_1 + e_2 + e_3)$$

## Level 2

1 point

What was the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Malayalam movies together?

☐

$$T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3) \quad T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3)$$

☐

$$200T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3) \quad 200T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3)$$

☐

$$T(m_1 m'_1, m_2 m'_2, m_3 m'_3) \quad T(m_1 m'_1, m_2 m'_2, m_3 m'_3)$$

☐

$$200T(m_1 m'_1, m_2 m'_2, m_3 m'_3) \quad 200T(m_1 m'_1, m_2 m'_2, m_3 m'_3)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3) \quad T(m_1 + m'_1, m_2 + m'_2, m_3 + m'_3)$$

1 point

Which of the following options are correct?

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$$T(x, y, z) = T(x, 0, 0) + T(0, y, 0) + T(0, 0, z) \quad T(x, y, z) = T(x, 0, 0) + T(0, y, 0) + T(0, 0, z)$$

☐

$$T(x, y, z) = T(x, y, 0) + T(0, 0, z) \quad T(x, y, z) = T(x, y, 0) + T(0, 0, z)$$

☐

$$T(x, y, z) = T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0) \quad T(x, y, z) = T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0)$$

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$$T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0) = T(x + y + z, 0, 0)$$

$$T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0) = T(x + y + z, 0, 0)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$T(x, y, z) = T(x, 0, 0) + T(0, y, 0) + T(0, 0, z) \quad T(x, y, z) = T(x, 0, 0) + T(0, y, 0) + T(0, 0, z)$$

$$T(x, y, z) = T(x, y, 0) + T(0, 0, z) \quad T(x, y, z) = T(x, y, 0) + T(0, 0, z)$$

$$T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0) = T(x + y + z, 0, 0)$$

$$T(x, 0, 0) + T(y, 0, 0) + T(z, 0, 0) = T(x + y + z, 0, 0)$$

1 point

Which of the following equalities correctly represent the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the Bengali and Hindi movie together?

☐

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), (h_3 + b_3)) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), (h_3 + b_3))$$

☐

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, 0, 0) + T(0, h_2 + b_2, 0) + T(0, h_3 + b_3, 0) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, 0, 0) + T(0, h_2 + b_2, 0) + T(0, h_3 + b_3, 0)$$

☐

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), 0) + T(0, 0, (h_3 + b_3)) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), 0) + T(0, 0, (h_3 + b_3))$$

☐

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, b_2, b_3) + T(0, h_2, h_3) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, b_2, b_3) + T(0, h_2, h_3)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), (h_3 + b_3)) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), (h_3 + b_3))$$

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, 0, 0) + T(0, h_2 + b_2, 0) + T(0, h_3 + b_3, 0) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, 0, 0) + T(0, h_2 + b_2, 0) + T(0, h_3 + b_3, 0)$$

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), 0) + T(0, 0, (h_3 + b_3)) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, (h_2 + b_2), 0) + T(0, 0, (h_3 + b_3))$$

$$T(0, b_2 + h_2, b_3 + h_3) = T(0, b_2, b_3) + T(0, h_2, h_3) \quad T(0, b_2 + h_2, b_3 + h_3) = T(0, b_2, b_3) + T(0, h_2, h_3)$$

1 point

Which of the following equality correctly represents the total amount of money the multiplex made in the specific day mentioned above by selling the tickets of the English movie?

☐

$$T(e_1, e_2, e_3) = 200(e_1 + e_2 + e_3) \quad T(e_1, e_2, e_3) = 200(e_1 + e_2 + e_3)$$

☐

$$T(e_1, e_2, e_3) = 250(e_1 + e_2 + e_3) \quad T(e_1, e_2, e_3) = 250(e_1 + e_2 + e_3)$$

☐

$$T(e_1, e_2, e_3) = 200e_1 + 250e_2 + 300e_3 \quad T(e_1, e_2, e_3) = 200e_1 + 250e_2 + 300e_3$$

☐

$$T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3) \quad T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3)$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3) \quad T(e_1, e_2, e_3) = 300(e_1 + e_2 + e_3)$$

[Check Answers](#)

Your score is: 0/8